E-WASTE & ENVIRONMENTAL MANAGEMENT

Case Studies

Japan, A Recycling-Oriented Society

INTRODUCTION:

Japan manages waste through proper waste collection, recycling, and modern technology. Its main aim is to reduce waste and save natural resources.

KEY FEATURES:

- Mottainai: Avoiding waste and respecting resources.
- Recycling Laws: Laws encourage recycling and reuse of materials.
- Waste Collection: Organized waste collection and recycling centres.
- Technology: Modern technology helps manage waste efficiently.

CIRCULAR ECONOMY BENEFITS AND LIMITATIONS:

- 3Rs: Reduce, Reuse and Recycle help decrease waste.
- Eco-Towns: Waste from one industry can be used as a resource by another.
- Innovation: Waste can be converted into ethanol for industrial use.
- Lessons for Other Countries: People should participate in waste management, follow rules, separate waste properly, and encourage companies to recycle.

SWOT ANALYSIS:

Strengths	Weaknesses
Strong laws, responsible citizens, and modern facilities.	High costs and heavy dependence on waste burning. Limited landfill space.
Opportunities	**Threats**
AI-based sorting, smart bins, and better recycling methods.	Air pollution from waste burning and difficulty recycling mixed plastic waste.



Mumbai: Bridging Informal and Formal E-Waste Management

INTRODUCTION:

Mumbai manages e-waste through households, local scrap dealers (kabadiwalas), recycling companies, manufacturers, government bodies, and social organizations. These groups work together to collect and recycle electronic waste.

E-waste stakeholders and their roles

Stakeholder	Main role
Consumers	Discard old electronics and decide where to give them.
Informal Collectors	Collect and sort e-waste and earn a living.
Authorized Recyclers	Safely recycle e-waste and recover useful materials.
Producers and Authorities	Make sure e-waste is collected and recycling rules are followed.
NGOs and Community Groups	Spread awareness and connect people with recycling services.

E-waste collection: who collects, and at what cost

Collection estimates



Benefits and risks

Benefits	Risks
Easy door-to-door collection.	Workers lack safety equipment.
Provides jobs and income.	Exposure to harmful substances.
Recovers copper and other useful materials.	Burning and chemicals pollute air, soil and water.
Strong local collection networks.	Unsafe recycling and poor tracking of waste.

Informal-to-formal integration



- Train and register informal waste collectors.
- Connect collectors with authorized recyclers and offer fair prices.
- Make e-waste collection easy and spread awareness about safe recycling.

SWOT Analysis: E-waste Management

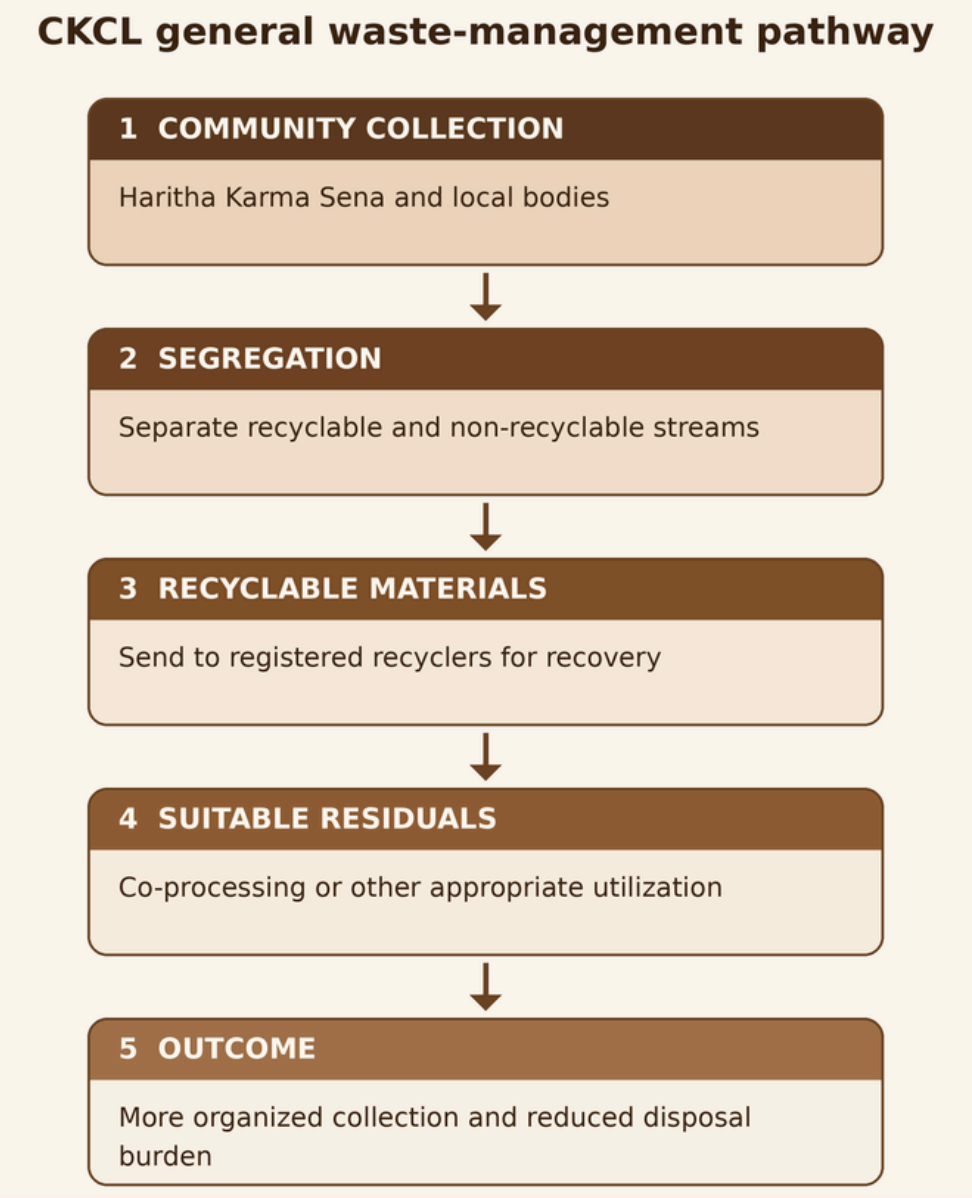


Kerala: Clean Kerala Company (CKCL)

INTRODUCTION:

Kerala manages non-biodegradable and old waste through a community-based system.

- Collection: Local bodies and community groups collect waste.
- Haritha Karma Sena: Helps collect waste from households.
- Segregation: Waste is separated, and recyclable materials are sent to registered recyclers.
- Waste Treatment: Non-recyclable waste is used for other purposes when possible.



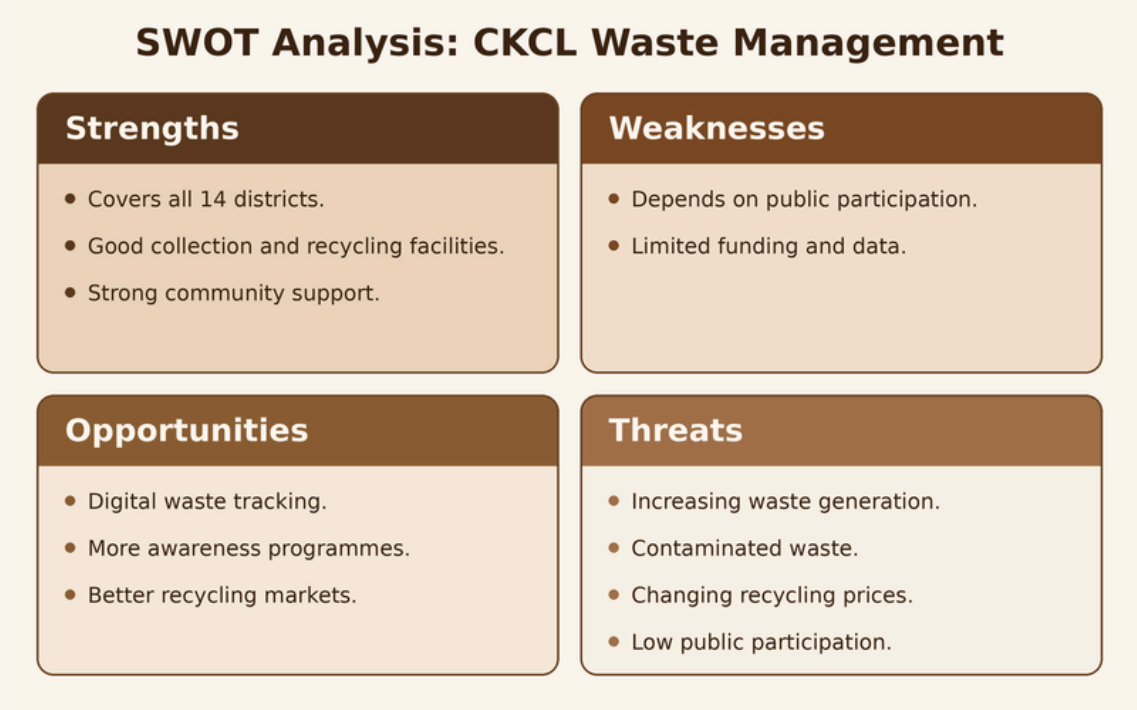
INFRASTRUCTURE:

- Districts Covered: 14
- Collection Centres (MCFs): 180+
- Recovery Centres (RRFs): 60+
- Monthly Processing: 15,000+ tonnes

Note: These figures are from the supplied report and are not independently verified.

Benefits and Lessons:

- Organized collection helps manage and separate waste properly.
- Recycling saves resources and reduces waste.
- Other states can adopt this model through local bodies, collection centres, awareness and digital tracking.
- E-waste should be sent to authorized recyclers for safe recycling.



Switzerland: Producer-Funded E-Waste Take-Back

INTRODUCTION:

- Switzerland manages e-waste through an organized collection and recycling system.
- Advance Recycling Fee (ARF): A small fee included in the purchase price pays for future recycling.
- Collection: Shops and collection centres accept old electronics for free.
- Producer Responsibility: Producers help manage e-waste through recycling organizations.
- Safe Recycling: Special facilities, inspections and certifications ensure proper recycling.

PERFORMANCE:

- More than 6,600 collection points.
- Around 85–95% of major electronic waste is formally collected.
- Easy collection and proper funding encourage safe disposal.

How the system works

1 PURCHASE

ARF included in the purchase price

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2 FINANCING AND COORDINATION

Funds support organized collection and recycling

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3 CONSUMER RETURN

Old electronics returned through retailers or collection points

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4 CERTIFIED PROCESSING

Hazardous components removed; materials recovered

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5 TRACEABILITY

Auditing helps monitor material flows

Key features and lessons

Feature	Lesson
Recycling Fee (ARF)	Ensure regular funding.
Collection Points	Make e-waste returns easy.
Special Facilities	Use authorized recyclers.
Certification and Audits	Track and check recycling activities.

SWOT Analysis: E-waste Recycling System

Strengths

- Prepaid recycling fees.
- Easy returns at shops.
- Many collection points.
- Certified recycling facilities.

Weaknesses

- People store old devices at home.
- Privacy concerns about phones.
- Limited reuse of devices.

Opportunities

- Repair and reuse electronics.
- Secure data wiping.
- Recover valuable metals.

Threats

- Increasing e-waste.
- Illegal dumping and exports.
- Difficult-to-repair products.

Samsung: Trade-In, Refurbishment and Recycling

INTRODUCTION:

- Samsung encourages customers to return old devices through its trade-in programme. Depending on their condition, devices are either repaired and reused or sent for recycling.

STRATERGIES:

- Reuse: Extend device life through repair, trade-in and refurbishment.
- Recycling: Recover useful parts and materials from old devices.
- Material Recovery: Reuse recovered materials, such as cobalt, in new batteries.
- India Context: Samsung follows India's E-Waste Management Rules, 2022 and EPR system.

Reported Figures

- Collection Systems: 100+ countries (2025).
- E-Waste Collected: 7.54 million tonnes by 2025.
- 2030 Target: 10 million tonnes.
- 2050 Target: 25 million tonnes.

COllection Growth:

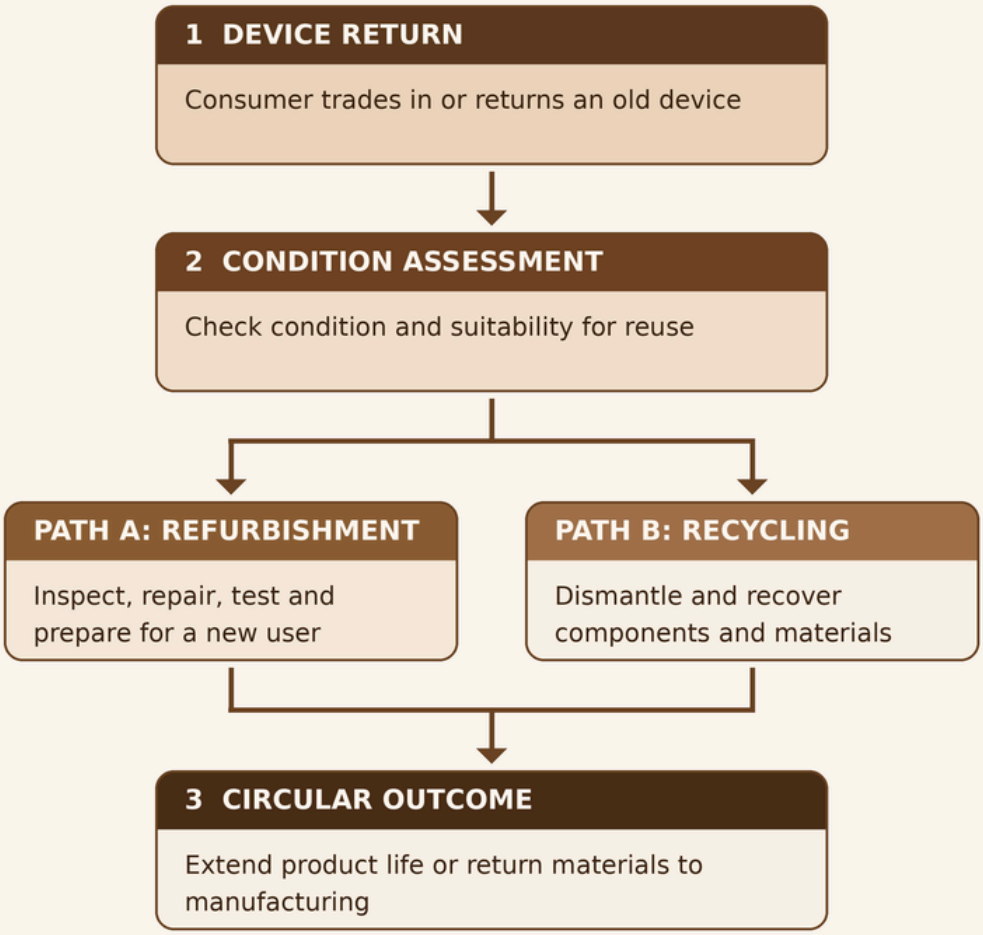
- Collection Growth
- 2023: 6.30 million tonnes.
- 2024: 6.91 million tonnes.
- 2025: 7.54 million tonnes.

Note: These figures and targets are reported by Samsung. The amount of waste reused, refurbished or recycled is not specified.

RECOMMENDATIONS:

- Report recycling, reuse and recovery rates clearly.
- Encourage repair and reuse before recycling.
- Provide easy collection and secure data deletion.
- Train informal collectors and connect them with authorized recyclers.
- Consumers should delete personal data and return devices through authorized channels.

What happens after trade-in?



SWOT Analysis: Device Trade-in Programme

Strengths

- Global reach and easy device returns.
- Reuse, refurbishment and material recovery.

Weaknesses

- Trade-in does not guarantee reuse.
- Limited recycling details.

Opportunities

- Growing demand for refurbished devices.
- Recovery of valuable metals.

Threats

- Privacy concerns.
- Lack of proper recycling facilities.
- Competition from informal recyclers.

Case studies: strengths and lessons

Case study	Main strength	Key lesson
Japan	Strong laws and organized waste management.	Combine public participation, laws and recycling.
Mumbai	Large informal collection network.	Connect collectors with authorized recyclers.
Kerala (CKCL)	Community collection system.	Improve collection and track performance.
Switzerland	Easy collection and advance recycling fees.	Make recycling convenient and affordable.
Samsung	Trade-in, repair and recycling.	Measure reuse and material recovery.

Key Takeaway

- Effective e-waste management requires easy collection, clear rules, safe recycling and proper tracking. Repair and reuse devices whenever possible, and recycle them when reuse is no longer practical.